VLSI Internship – Task 1 Report

Introduction to VLSI Design Flow and Basic Digital Logic Implementation

Name: T. Ruchitha
Institution: RNS Institute of Technology (RNSIT), Bangalore
Internship: VLSI (Very Large Scale Integration) – Maincrafts Technology
Task: Task 1 – Introduction to VLSI Design Flow and Basic Digital Logic Implementation

1. VLSI Design Flow

VLSI (Very Large Scale Integration) is the process of integrating millions to billions of transistors onto a single silicon chip, used in microprocessors, mobile SoCs, memory chips, and AI hardware.

The standard VLSI design flow followed in industry consists of the following stages:

- 1. System Specification – Defining chip functionality, performance, power, and area constraints.
- 2. Architecture Design – High-level block diagram and functional data flow.
- 3. RTL Design – Coding the design in Verilog/VHDL at Register Transfer Level.
- 4. Functional Verification – Simulating the RTL to check correctness.
- 5. Synthesis – Converting RTL code into a gate-level netlist.
- 6. Physical Design – Floorplanning, placement, and routing.
- 7. Timing & Power Analysis – Ensuring the design meets speed and power targets.
- 8. Fabrication – Manufacturing the chip on silicon.
- 9. Testing & Validation – Verifying the fabricated chip works as intended.

Task 1 focuses on Steps 1 and 2 (System Specification and Architecture Design) and builds the digital logic foundation required before RTL coding.

2. Digital Logic Fundamentals – Truth Tables

AND Gate (Output = 1 only if all inputs are 1)

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

OR Gate (Output = 1 if any input is 1)

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

NAND Gate (Inverted AND)

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

NOR Gate (Inverted OR)

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

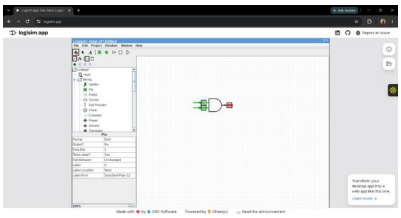
XOR Gate (Output = 1 if inputs are different)

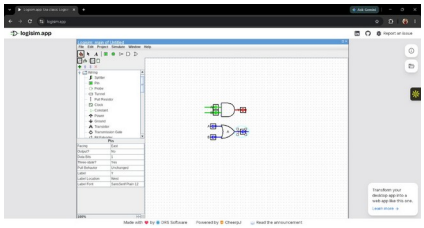
A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

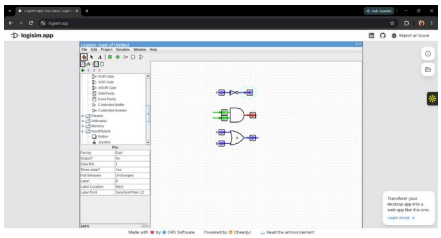
NOT Gate (Inverts the input): Input 0 → Output 1, Input 1 → Output 0.

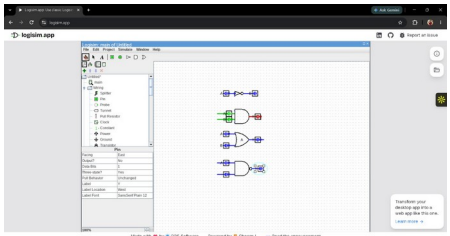
3. Logic Gate Simulations (Logisim)

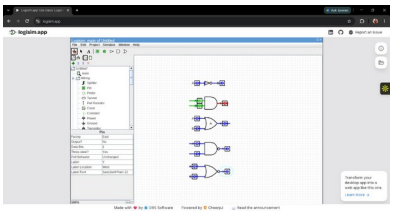
The following gates were built and simulated in Logisim, verifying correct output for all input combinations.

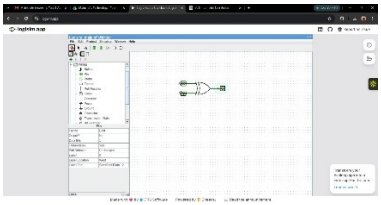








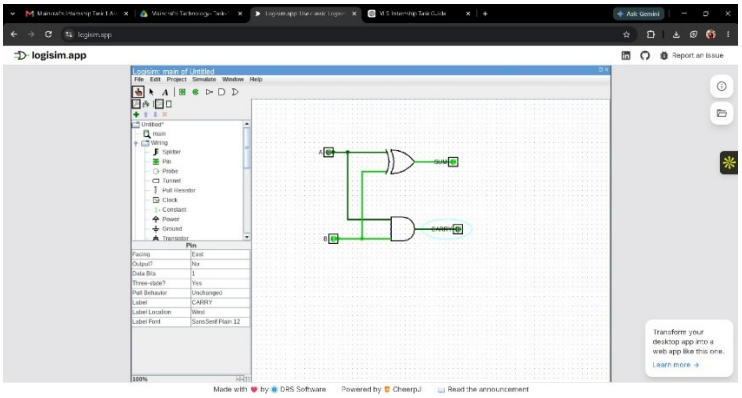




4. Combinational Circuit – Half Adder

A Half Adder was designed using logic gates, with:

- Sum = A XOR B
- Carry = A AND B



4a. Combinational Circuit – Full Adder (Optional)

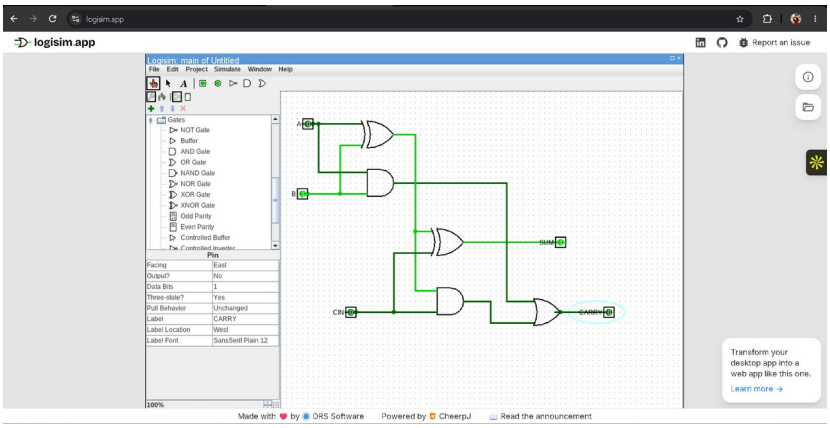
A Full Adder extends the Half Adder by adding a Carry-in (Cin), and can be built using two Half Adders and an OR gate:

- Sum = A XOR B XOR Cin
- Carry-out = (A AND B) OR (Cin AND (A XOR B))

Full Adder Truth Table

A	B	Cin	Sum	Cout
0	0	0	0	0

0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



5. Observations

Inputs used: All combinations of A and B (0,0 / 0,1 / 1,0 / 1,1) were tested for each gate and the Half Adder.
Output behavior: All gates and the Half Adder produced outputs matching their respective truth tables.
Mistakes and corrections: [Add any wiring/connection errors you fixed while simulating, if any.]

6. Key Learning Outcomes

- Explained what VLSI is and where it is used
- Described the complete VLSI design flow
- Understood digital logic fundamentals (gates, Boolean algebra, truth tables)
- Built and simulated basic logic circuits in Logisim
- Prepared foundational knowledge for RTL and Verilog-based tasks ahead